

DroneArjuna (DA)

Gap Resolution Plan

Open items only — cross-referenced against DA-Project-Timeline-AI-Adjusted

ACS Technologies Limited

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Classification: RESTRICTED / CONFIDENTIAL

Cross-referenced against:

DA-Project-Timeline-AI-Adjusted (Phase 1–6 task register)

1. Purpose and Scope

This document is restricted to **open gaps only**. It does not repeat completed work. All gap items are sourced from the *DA As-Built Implementation Status Report (5 August 2026)* and mapped to task identifiers and AI-adjusted effort estimates from the *DA-Project-Timeline-AI-Adjusted* document dated 6 June 2026.

Phases 0, 1, and 2 are fully complete and are excluded from this report. Gaps exist across Phases 3, 4, 5, and 6. The total remaining AI-adjusted effort is approximately **6.5–7.5 calendar weeks** (human baseline: ~30–38 weeks).

Phase	Status	Open Tasks	AI-Adjusted Remaining	Human Baseline
Phase 3 — Advanced Flight + Visualisation	Partial	4 tasks (P3-01 to P3-04)	~10–12 days	8–10 weeks
Phase 4 — Drone Inventory	Partial	8 tasks (P4-01 to P4-08)	~7–9 days	6–8 weeks
Phase 5 — Drone Analyst (AI/CV)	Scaffold only	10 tasks (P5-02 to P5-11)	~12–14 days	10–12 weeks
Phase 6 — Hardening / Security / UX	Partial	7 tasks (P6-01, P6-05–P6-11)	~5–6 days	6–8 weeks
TOTAL		29 open tasks	~6.5–7.5 weeks	~30–38 weeks

Effort symbols (from AI-Adjusted Timeline): XS = 15–30 min | S = 1–2 h | M = 3–6 h | L = 1–2 days | XL = 3–5 days | ■ (fixed) = human-gated, no AI speedup

2. Phase 3 — Advanced Flight + Visualisation Gaps

Phase total remaining: 10–12 calendar days (human baseline: 8–10 weeks). Geofence enforcement, auto-RTL, multi-drone deconfliction, and vessel feed integration are already implemented. The sole gap is the **3D visualisation suite** and the **telemetry replay player**, both dependent on Cesium.js.

Cesium.js is flagged as the dominant cost for this phase — a 600k-line library with its own rendering paradigm. All four tasks below depend on P3-01 completing first.

#	Task	AI Effort	Dependency	Notes
P3-01	Cesium.js integration — scaffold Map3DCanvas.tsx	XL (3–4 days)	None	Largest Phase 3 task; large novel library, expensive
P3-02	3D terrain draping for mission routes	L (1.5 days)	P3-01	Overlay mission waypoint paths on 3D terrain
P3-03	3D live drone tracking (glTF model + telemetry)	M (4 h)	P3-01	Real-time drone position driven from existing gRPC
P3-04	Telemetry replay player (timeline scrubber)	M (4 h)	None (uses existing telemetry history endpoint)	Symbolic timeline; API endpoint already exists

What is already done in Phase 3:

- Geofence enforcement — breach detection, event-driven RabbitMQ consumer, auto-RTL trigger
- Multi-drone airspace deconfliction — constraint-programming fleet assignment solver (+ experimental quantum solver)
- Vessel-relative home point — live update from HF position feed
- HF position feed end-to-end — ingests vessel GPS, updates naval vessel records
- 2D map workspace (Fly + Plan) with geofence overlays, mission route rendering, regulatory-zone awareness
- pytest advanced flight suite — mission, geofence, deconfliction logic covered

Sprint guidance (from AI-Adjusted Timeline, Weeks 4–6): Begin P3-01 Cesium scaffold in Weeks 4–5 (allow full 4 days). Run P3-02 and P3-03 in Weeks 5–6 after scaffold is stable. P3-04 (replay player) can be parallelised with P3-02/P3-03 as it has no Cesium dependency.

3. Phase 4 — Drone Inventory (Knowledge Base) Gaps

Phase total remaining: 7–9 calendar days (human baseline: 6–8 weeks). The backend inventory module is implemented as a first-version wrapper over Drone Master data. The search engine is deployed and indexed at startup. What is missing is the **dedicated Inventory frontend workspace**, the **full Elasticsearch deep integration** replacing ILIKE queries, and the **Threat System / ORBAT knowledge-base page**.

#	Task	AI Effort	Dependency	Notes
P4-01	InventoryWorkspace.tsx — new 6th workspace shell	S (1.5 h)	None	Shell/router only; content filled by P4-02 onwards
P4-02	Drone catalogue browser — card grid + facet	M (3 h)	P4-01	Consumes existing /api/inventory/drones endpoint
P4-03	Drone spec detail page + performance chart	M (3 h)	P4-01	Consumes existing /api/inventory/drones/{id} endpoint
P4-04	Side-by-side comparison view	S (1.5 h)	P4-01	API comparison endpoint already exists
P4-05	Payload catalogue browser	M (2 h)	P4-01, P2-06	Depends on PayloadManager backend (complete)
P4-06	Elasticsearch integration — replace ILIKE queries	XL (3–4 days)	Running ES cluster	New infra wiring + index mapping redesign; dominant cost
P4-07	Instant search bar (debounced)	XS (30 min)	P4-06 or existing search endpoint	Frontend only; trivial once backend search is live
P4-08	Threat system / ORBAT knowledge-base page	L (2 days)	P4-01	Novel domain model; CRUD endpoints + DB mod

Note on Elasticsearch: The search engine is already deployed and indexed at startup (inventory data bulk-indexed on application boot). P4-06 is specifically about replacing the SQL ILIKE pattern-match queries with native Elasticsearch queries and redesigning the index mapping for faceted search. This is the dominant cost and should be scheduled separately from the UI tasks (P4-01–P4-05 and P4-07 can be completed in a single focused day).

Sprint guidance (from AI-Adjusted Timeline, Weeks 3 and 7–8): Complete P4-01 to P4-05 and P4-07 (all UI tasks) in Week 3 in parallel with Phase 2 wrap-up. Defer P4-06 (Elasticsearch deep integration) and P4-08 (ORBAT page) to Weeks 7–8 after the ML pipeline sprint.

4. Phase 5 — Drone Analyst (AI/CV Pipeline) Gaps

Phase total remaining: 12–14 calendar days (human baseline: 10–12 weeks). This is the largest gap area. The analyst module is a scaffold only: job registry, telemetry stats/time-series endpoints, model registry, and detection-result storage are implemented and backed by the database. **No object-detection inference, video-frame processing, or background-worker infrastructure was found.** Object storage (MinIO) is deployed but unused in code.

Fixed bottleneck: The GPU/ML environment (CUDA drivers, YOLOv8 model weights, Docker GPU passthrough) is human-gated and remains at original duration regardless of AI assistance. P5-03 Celery worker infrastructure is also partially human-gated (Docker Compose changes, broker config). These must be provisioned before P5-04/P5-05 can run.

#	Task	AI Effort	Dependency	Notes
P5-02	MinIO upload endpoint (POST /api/analyst/upload)	S (1.5 h)	MinIO deployed	Object storage deployed but unused; first integration
P5-03	Celery worker infrastructure — broker config, worker registry	L (1.5 days)	RabbitMQ/Redis broker	New infra; Docker Compose changes required
P5-04	YOLOv8 object detection Celery task	L (2 days)	P5-03 + GPU/CUDA env	ML model setup; GPU config is fixed-cost bottleneck
P5-05	Change detection worker (pixel-diff algorithm)	L (1.5 days)	P5-03	CPU-only; no GPU dependency
P5-06	Video frame extraction Celery task	M (4 h)	P5-03	ffmpeg-based frame sampling
P5-07	Analyst workspace UI — job submission form	M (3 h)	None (uses existing job API)	Job registry API already present
P5-08	Detection results viewer — bounding-box overlay	L (1.5 days)	P5-04	Complex canvas rendering; depends on detection output
P5-09	Change detection map layer in Plan/Fly workspace	M (3 h)	P5-05	Overlay change-mask rasters on existing 2D map
P5-10	Telemetry analytics dashboard — rich charts	M (3 h)	None	API already present; frontend only
P5-11	Automated telemetry report PDF export	L (2 days)	P5-10	Scheduled report generation; uses telemetry stats

What is already done in Phase 5:

- AnalysisJob model + DB migration — P5-01 is complete (job persistence is implemented)
- Job registry — create / list / detail / cancel endpoints implemented
- Model registry — list / detail endpoints implemented
- Detection-result storage — DB model and listing endpoint present
- Telemetry stats / time-series endpoints — fully implemented

Sprint guidance (from AI-Adjusted Timeline, Weeks 4–7): Begin P5-02, P5-07, and P5-10 in Weeks 4–5 (these have no infrastructure dependency). P5-03 Celery infrastructure runs in Week 6 after Phase 3 Cesium work. P5-04, P5-05, P5-06 follow in Weeks 6–7 once the worker scaffold is stable. P5-08, P5-09, P5-11 close out Phase 5 in Week 7.

5. Phase 6 — Hardening, Security and UX Polish Gaps

Phase total remaining: 5–6 calendar days (human baseline: 6–8 weeks). Core security controls are already in place: JWT auth with token blacklist, rate limiter (cache-backed), audit logger, input sanitiser, and password policy. The open items are test coverage verification, infrastructure hardening, accessibility, load testing, and CI pipeline.

#	Task	AI Effort	Notes / Fixed Costs
P6-01	Full pytest coverage pass — achieve >=80% line of code coverage	L (1.5 days)	~854 tests exist; coverage % not yet verified; gap-fill tests needed for frontend
P6-05	TLS/HTTPS nginx reverse proxy	S (1.5 h)	Infrastructure config; certificate provisioning may require human action
P6-06	WCAG 2.1 AA accessibility pass — focus rings, MR (4 h)	MR (4 h)	Requires human visual review of rendered UI; AI writes the fixes
P6-07	React error boundaries per workspace	XS (30 min)	App shell error boundary already present; per-workspace wrapping is additional
P6-08	Service worker + offline indicator	M (3 h)	PWA offline capability; Workbox or custom service worker
P6-09	Locust load test — 10 drones x telemetry + 3 concurrent missions	M (3 h)	Requires running system; baseline performance numbers needed before this
P6-10	Docker image hardening (non-root user, read-only filesystem)	S (1.5 h)	Dockerfile changes; verify USB passthrough still works under non-root
P6-11	GitHub Actions CI pipeline	M (3 h)	Lint + pytest + build on push; Docker layer caching recommended

What is already done in Phase 6:

- JWT token blacklist — cache-backed revocation (P6-02 complete)
- Rate limiting on auth — cache-backed sliding window (P6-03 complete)
- Input sanitisation — control-character stripping, HTML escaping, injection-pattern logging (P6-04 complete)
- Password policy — length/complexity/blocklist enforcement (partial P6-04 complete)
- Audit logger — append-only audit log for sensitive operations (complete)

Sprint guidance (from AI-Adjusted Timeline, Weeks 8–9): Run P6-01 through P6-11 as a single hardening sprint. P6-01 (coverage) is the longest task; run it first. P6-06 (WCAG) requires a human visual review pass — schedule alongside a UI walkthrough session. P6-09 (load test) should run on a deployed environment with all phases active.

6. Fixed-Cost Bottlenecks — AI Does Not Accelerate These

The following items remain at their original human-baseline duration regardless of AI coding assistance. They are the true constraints on delivery velocity and must be provisioned or planned in advance.

Bottleneck	Affects	Why It Cannot Be Accelerated	Action Required
Cesium.js library integration	P3-01 to P3-03	600k-line library with its own rendering paradigm; AI repeats a full day of tasks of Abhishek	Allocate a full day of tasks of Abhishek to other Phase 3
GPU / ML environment (CUDA, Model weights)	P4-04	YOLOv8 requires CUDA drivers, model weights download, Docker, P4 phase and get started through before	Download Docker, P4 phase and get started through before
Elasticsearch index redesign	P4-06	New index mapping design, data migration run, and stand up new Elasticsearch environment with new design	Stand up new Elasticsearch environment with new design
WCAG visual accessibility review	P6-06	Must be done by a human inspecting the rendered code changes	Schedule a human to review code changes
Load / SITL testing	P6-09, P1-03	Requires a fully deployed running system; test output depends on real time stage network behavior	Run tests on real time stage network behavior
Security audit / penetration testing	Phase 6	Threat modelling and manual code review beyond AI capabilities	Engage security review activities P6-01 to P6-11 sprint is complete

7. Revised Gap Sprint Plan

The following sprint plan covers only the remaining gap items, sequenced per the AI-Adjusted Timeline (from Week 3 onward — Weeks 1–2 closed Phases 1 and 2). Effort estimates use the AI-adjusted scale.

Week 3 — Phase 2 Completion + Phase 4 UI

- Mon: P2-16 (simulation preview animation), P2-17 (pytest flight suite) → Phase 2 fully closed
- Tue–Fri: P4-01 InventoryWorkspace tab, P4-02 catalogue browser, P4-03 spec detail page, P4-04 comparison view, P4-05 payload browser, P4-07 instant search bar
- Milestone: Phase 4 UI closed (Elasticsearch P4-06 and ORBAT P4-08 deferred to Weeks 7–8)

Milestone: Phase 2 complete | Phase 4 UI complete

Weeks 4–5 — Cesium Scaffold + Analyst Foundation

- P3-01: Cesium.js scaffold Map3DCanvas.tsx — allow full 4 days; this is the largest remaining task
- Parallel (no Cesium dependency): P5-02 MinIO upload endpoint, P5-07 analyst workspace UI, P5-10 telemetry analytics dashboard
- Note: P5-02/P5-07/P5-10 can be completed in ~1 day combined

Milestone: Cesium scaffold stable | Analyst UI foundation ready

Weeks 5–6 — Phase 3 Completion

- P3-02: 3D terrain draping for mission routes (1.5 days)
- P3-03: 3D live drone tracking — glTF model + telemetry feed (4 h)
- P3-04: Telemetry replay player with timeline scrubber (4 h) — can run parallel to P3-02
- P3-09: pytest advanced flight suite update (1.5 h)

Milestone: Phase 3 fully closed

Weeks 6–7 — Phase 5 ML Pipeline

- P5-03: Celery worker infrastructure — broker config, worker/ directory, Docker Compose (1.5 days)
- P5-04: YOLOv8 object detection Celery task (2 days) — GPU environment must be provisioned first
- P5-05: Change detection worker / pixel-diff algorithm (1.5 days)
- P5-06: Video frame extraction Celery task (4 h)
- P5-08: Detection results viewer with bounding-box overlay (1.5 days)
- P5-09: Change detection map layer in Plan/Fly workspaces (3 h)
- P5-11: Automated telemetry report PDF export (2 days)

Milestone: Phase 5 fully closed

Weeks 7–8 — Phase 4 Completion

- P4-06: Elasticsearch integration — replace ILIKE, redesign index mapping (3–4 days)
- P4-08: Threat system / ORBAT model + knowledge-base page (2 days)

Milestone: Phase 4 fully closed

Weeks 8–9 — Phase 6 Hardening Sprint

- P6-01: Full pytest coverage pass, achieve $\geq 80\%$ line coverage (1.5 days)
- P6-05: TLS/HTTPS nginx reverse proxy (1.5 h)
- P6-06: WCAG 2.1 AA pass — focus rings, ARIA, contrast (4 h + human review)
- P6-07: React error boundaries per workspace (30 min)
- P6-08: Service worker + offline indicator (3 h)
- P6-09: Locust load test — 10 drones x telemetry + 3 missions (3 h on staging)
- P6-10: Docker image hardening — non-root, read-only fs (1.5 h)
- P6-11: GitHub Actions CI pipeline (3 h)

Milestone: Phase 6 fully closed

Weeks 9–10 — Buffer + Integration Testing

- Buffer for integration issues across all phases active simultaneously
- Final QA on real SITL with Phase 3 Cesium, Phase 5 ML workers, and Phase 6 hardening live
- DA-STD / DA-STR baseline document update

Milestone: PROJECT COMPLETE

8. Consolidated Open-Task Register

All 29 open tasks listed below in phase order. Tasks marked with a fixed bottleneck symbol (■) have a human-gated component that AI cannot accelerate.

#	Phase	Task Description	AI Effort	Fixed Cost?
P3-01	Phase 3	Cesium.js integration — scaffold Map3DCanvas.tsx	XL (3–4 d)	Yes — library complexity
P3-02	Phase 3	3D terrain draping for mission routes	L (1.5 d)	—
P3-03	Phase 3	3D live drone tracking (glTF + telemetry feed)	M (4 h)	—
P3-04	Phase 3	Telemetry replay player (timeline scrubber)	M (4 h)	—
P4-01	Phase 4	InventoryWorkspace.tsx — new 6th workspace tab	S (1.5 h)	—
P4-02	Phase 4	Drone catalogue browser — card grid + facet filters	M (3 h)	—
P4-03	Phase 4	Drone spec detail page + performance chart	M (3 h)	—
P4-04	Phase 4	Side-by-side comparison view	S (1.5 h)	—
P4-05	Phase 4	Payload catalogue browser	M (2 h)	—
P4-06	Phase 4	Elasticsearch integration — replace ILIKE queries	XL (3–4 d)	Yes — cluster + index design
P4-07	Phase 4	Instant search bar (debounced)	XS (30 m)	—
P4-08	Phase 4	Threat system / ORBAT knowledge-base page	L (2 d)	—
P5-02	Phase 5	MinIO upload endpoint (POST /api/analyst/upload)	S (1.5 h)	—
P5-03	Phase 5	Celery worker infrastructure — broker config, worker/ dir	L (1.5 d)	Partial — Docker config
P5-04	Phase 5	YOLOv8 object detection Celery task	L (2 d)	Yes — GPU/CUDA env
P5-05	Phase 5	Change detection worker (pixel-diff algorithm)	L (1.5 d)	—
P5-06	Phase 5	Video frame extraction Celery task	M (4 h)	—
P5-07	Phase 5	Analyst workspace UI — job submission form	M (3 h)	—
P5-08	Phase 5	Detection results viewer — bounding-box overlay	L (1.5 d)	—
P5-09	Phase 5	Change detection map layer in Plan/Fly workspaces	M (3 h)	—
P5-10	Phase 5	Telemetry analytics dashboard — rich charts	M (3 h)	—
P5-11	Phase 5	Automated telemetry report PDF export	L (2 d)	—
P6-01	Phase 6	Full pytest coverage pass — >=80% line coverage	L (1.5 d)	—
P6-05	Phase 6	TLS/HTTPS nginx reverse proxy	S (1.5 h)	Partial — certificate
P6-06	Phase 6	WCAG 2.1 AA pass — focus, ARIA, contrast	M (4 h)	Yes — human visual review
P6-07	Phase 6	React error boundaries per workspace	XS (30 m)	—
P6-08	Phase 6	Service worker + offline indicator	M (3 h)	—
P6-09	Phase 6	Locust load test — 10 drones x telemetry + 3 missions	M (3 h)	Yes — running system
P6-10	Phase 6	Docker image hardening (non-root, read-only fs)	S (1.5 h)	—
P6-11	Phase 6	GitHub Actions CI pipeline	M (3 h)	—

This report was generated on 10 August 2026 from the DA-Implementation-Status report (5 August 2026) cross-referenced with the DA-Project-Timeline-AI-Adjusted document (6 June 2026). It should be regenerated as phases close.